

Your Task 2 notebook is actually quite strong already. It is well-structured, analytical, and much closer to HD level than Task 1.

You already covered:

- EDA
- skewness analysis
- scale comparison
- correlation analysis
- standardisation
- min-max scaling
- log transformation
- power transformation
- Box-Cox transformation
- regression diagnostics
- final recommendations
- conclusion

That aligns very well with the assignment requirements and rubric.

But there are still a few important improvements that would strengthen it further and make it safer for HD.

What is GOOD already

Your notebook already has:

- ✓ clear sectioning
- ✓ proper explanations
- ✓ justification of methods
- ✓ before-vs-after comparison
- ✓ skewness evaluation
- ✓ linear regression diagnostics
- ✓ final recommendations per feature
- ✓ strong narrative flow

This directly matches the rubric's HD descriptors:

“Different methods are well analysed; effects shown in detail.”

What you should IMPROVE

1. Add QQ Plots (Very Important HD Improvement)

Right now you analyse skewness numerically, but you do not visually check normality.

Adding QQ plots makes the notebook look much more statistically rigorous.

Add this after Section 3.2 Distribution & Skewness

Insert a markdown cell:

3.2.1 QQ Plot Analysis

QQ plots are used to visually assess whether the variables approximately follow a normal distribution. Strong deviations from the diagonal line indicate skewness or heavy tails.

Then add this code:

```
from scipy.stats import probplot

# QQ plots for raw features

for col in features:

    plt.figure(figsize=(5,5))

    probplot(raw[col], dist="norm", plot=plt)

    plt.title(f'QQ Plot - {col}')

    plt.show()
```

2. Add Residual Plot for Regression Diagnostics

You already compute R^2 .

But the rubric mentions checking linear regression suitability.

Residual plots make your justification much stronger.

Add this after your regression diagnostics section (after Cell 37/38)

Markdown:

5.2 Residual Analysis

Residual plots help assess whether the transformed data better satisfies the assumptions of linear regression, particularly homoscedasticity and linearity.

Code:

```
# Residual plot after Box-Cox transformation
```

```
residuals = y_test_bc - pred_bc
```

```
plt.figure(figsize=(7,5))
```

```
plt.scatter(pred_bc, residuals)
```

```
plt.axhline(y=0, linestyle='--')
```

```
plt.xlabel("Predicted Values")
```

```
plt.ylabel("Residuals")
```

```
plt.title("Residual Plot - Box-Cox Transformed Data")
```

```
plt.show()
```

3. Explain WHY Municipality Was Excluded

You mention it briefly, but for HD you should justify categorical exclusion more clearly.

Add this under Section 2

`municipality` was excluded from the transformation analysis because it is a categorical attribute rather than a continuous numeric feature. Since this task focuses on scaling and reshaping methods for numerical variables in preparation for linear regression, only continuous numeric predictors were considered.

4. Add Small Comparison Table of Final Recommendations

This is a BIG presentation booster.

Add this before the conclusion.

Markdown

6.2 Final Transformation Summary Table

Code

```
recommendation_table = pd.DataFrame({
```

```
    'Column': [
        'number_of_houses',
        'number_of_units',
        'population',
        'aus_born_perc',
        'median_income',
        'median_house_price'
    ],
```

```
    'Recommended Transformation': [
        'Box-Cox',
        'Box-Cox',
        'Box-Cox',
        'None / Standardisation only',
        'Box-Cox',
        'Box-Cox'
    ],
```

```
'Reason': [  
    'Strong positive skew',  
    'Very strong positive skew',  
    'Moderate positive skew',  
    'Already near symmetric',  
    'Improved linearity',  
    'Reduce variance and improve regression assumptions'  
]  
})
```

recommendation_table

This makes your final recommendation section much cleaner and more professional.

5. Add One Final HD-Level Reflection Sentence

At the end of the conclusion, add:

Overall, Box-Cox transformation produced the most statistically suitable feature set for future linear regression modelling because it simultaneously improved feature symmetry, reduced skewness, stabilised variance, and modestly improved predictive performance.

MOST IMPORTANT THING

Your notebook is already good enough for roughly:

- Distinction / High Distinction borderline

These improvements mainly help with:

- HD polish
- presentation
- methodology justification
- visual evidence
- regression assumption discussion

which the rubric heavily rewards.

